“Grain3D” - Spatial Granular Synthesizer

By working with multichannel audio in different surroundings over the past 2 years, I made myself familiar with the common multichannel audio production workflows. For example, with the “IEM PlugIn Suite” which often handles all the involved processes of the spatialization for binaural/ambisonic purposes. In my Bachelor-Thesis I also used another approach which is called “distance based amplitude panning” which allows Spatialization for irregular speaker setups. However, the process of the actual spatialization of the virtual sound sources stays the same - audio channels from a single composition channel or a single instrument input channel are handled as a one virtual sound source. This source is then placed and possibly set in motion in a virtualized space, which represents the actual speaker arrangement in real space.

In the case of granular synthesis this approach is a bit cumbersome, because the sound produced by such an instrument is the actual sum of many individual sample playback instances („grains“). Therefore it would be nice to handle the sound produced by a single grain as a single virtual sound source, which then can be placed at different spatial locations. A common practice in traditional stereo granular synthesizers is that the position in the stereo field varies per grain. In “Grain3d” that approach should be taken but be applied to the spatial 3D field, so that each grain contains information about its position in the virtual three-dimensional space.

The spatialization process in „Grain3D“ therefore needs to happen at the stage of synthesis and not as a post processing effect via external spatialization tools. So, for the spatialization algorithm I plan to go with „Distance Based Amplitude Panning“ or „Vector Based Amplitude Panning“ rather then with an Ambisonics approach as the maths for that seems to be way more complex and thus less flexible in terms of user application & performance scenarios with irregular speaker setups.

Very similar approaches (in two-dimensional space) can be found in a variety of existing tools and instruments. For example the Max4Live Plugin „Vector Grain“ by Dillon Bastan (X or Y position modulates stereo panning) or the VST Plugin „Dust“ by Soundmorph (XY-Position modulates binaural panning). For generating motion, both examples utilize an approach, which is inspired by computer graphics - particle systems and vector maths for positional movement and behaviour. My plan is to also apply this approach. Furthermore, also to enable sound sources to be set in motion by complex motion patterns, like simulated physical motion or procedurally generated paths. I’d like to use Houdini for that. So, there is the need to build a JSON File importer to read the baked position data, generated in Houdini. The following graph shows a rough concept of the possible software structure.

